

Software Requirements Specification

PropManage BW — Property Management System

67

Functional requirements

46

Non-functional

18

Use cases

45

Pages

THE PROBLEM

Why PropManage BW exists

How landlords work today

- Rent tracked in spreadsheets and notebooks
- Arrears found at month end, not when they arise
- Repairs reported by phone, with no record
- Receipts written by hand, or not at all

Small and medium landlords — the target market — have no system of record.

1

One accurate record

Properties, units, tenants and leases in a single place

2

Rent raised automatically

No charge is forgotten, every month, without human effort

3

Arrears visible immediately

Overdue rent surfaces the day it arises, not at month end

Scope — bounded in both directions

In scope

- Accounts and role-based access for three roles
- Guided onboarding: property, houses, units, tenants
- Portfolio: properties, units and photographs
- Tenants, leases and unit assignment
- Automatic rent cycle, payments and receipts
- Maintenance workflow and tenancy messaging
- Dashboards and spreadsheet exports

Out of scope

- Online payment gateway or mobile money
- Automated SMS and e-mail reminders
- Subscription billing for the price plans
- Lease documents and e-signature
- Tenant credit screening
- A native mobile application

Everything excluded is either specified as future scope (FS-01...FS-09) or given a reason.

SECTION 2.3

Four user classes, one system

Landlord	Tenant	Administrator	Prospective customer
Up to 500	Up to 5,000	1 - 3	Unbounded
expected users	expected users	expected users	expected users
Laptop and mobile	Mobile phone	Desktop	Mobile or desktop
Knows what rent is outstanding; records payments; resolves repairs	Checks rent status, gets a receipt, reports a fault	Monitors adoption and collections across all landlords	Understands what the product does and what it costs

The tenant class is the largest and mostly on a phone — which is why the mobile layout requirements are the strictest in the document.

SECTION 3

What the specification contains

67

**Functional
requirements**

FR-01 ... FR-67

46

**Non-functional
requirements**

NFR-01 ... NFR-46

24

**External interface
requirements**

UI · HW · SW · CI

18

Use cases

12 specified in full

Seven quality attributes

Performance · security · usability · reliability and availability · maintainability · scalability
· compatibility

Verified against the build

Validation limits, enumerated values and scheduled behaviour taken from the
implemented system

The automated rent cycle

The heart of the system: rent is raised without anyone doing anything.



Why it matters

The cycle is idempotent (NFR-33) — running it any number of times in a month produces exactly one charge per tenant. A retry after a failure can never double-charge a tenant.

Payments, arrears and receipts

Payment status

pending

due date passes

overdue

payment recorded

paid

terminal state

Rent is received outside the system — bank transfer, cash or mobile money — and recorded afterwards. The system is a record of payment, not a means of payment (ASM-01).

Rules that protect the money

FR-33

A payment already marked paid cannot be recorded again — a double entry cannot understate arrears

FR-36

A receipt is issued only for a payment whose status is paid

FR-37

Only the managing landlord or the tenant named on the payment may obtain it

FR-32

A payment date that is not a valid date is rejected before anything is written

Authorisation lives in the database

Every table carries row-level rules. A defect in the interface cannot, on its own, expose one landlord's portfolio to another.

NFR-10

Deny by default

Row-level security is enabled on every table holding landlord, tenant, payment, maintenance or message data. A row is readable only where a policy grants it.

NFR-11

Portfolio isolation

A landlord reads and modifies only their own records. A request for another landlord's data returns nothing and changes nothing — even when it is well formed and the identifier is valid.

NFR-13

Decided on the server

No authorisation decision depends on a value from the browser. Changing client-side state cannot widen what a user can reach.

Tenant data is personal data. Botswana's Data Protection Act 2018 is recorded as a constraint (CON-10), not an afterthought.

QUALITY OF THE REQUIREMENTS

Measurable, not vague

Every non-functional requirement states a criterion that can be tested rather than argued about.

AVOIDED

"The system should be fast"

"It should be user-friendly"

"It should be reliable"

"It should be accessible"

SPECIFIED INSTEAD



NFR-01

The dashboard becomes interactive within 2.5 seconds at the 95th percentile, for up to 20 properties and 200 units



NFR-23

A new landlord completes onboarding in 10 minutes without training — 4 of 5 test participants succeed



NFR-29

99.5% availability each calendar month, outside announced maintenance



NFR-24

Contrast of at least 4.5:1 for normal text, meeting WCAG 2.1 level AA

SECTION 3.5

What the SRS deliberately does not promise

Nine capabilities are intended but excluded from this baseline. They carry their own identifiers so they are never mistaken for acceptance criteria.

FS-01

Online payment gateway and mobile money

FS-02

Automated rent reminders by e-mail and SMS

FS-03

Subscription billing and plan limits

FS-04

Password change and two-factor authentication

FS-05

Lease documents and renewal alerts

FS-06

Filtering and search on the maintenance board

An honest specification

Three screens already show controls that are not connected to anything — the password fields, the notification preferences and the maintenance filters. The SRS records them as future scope rather than specifying them as delivered behaviour, so the document cannot overstate what the system does.

IN CLOSING

Every requirement is traced and verifiable

The traceability matrix maps each requirement to the feature it realises, the use case that exercises it, and the method that will confirm it.

Test

Executed against a defined
expected result

Demonstration

Operated and observed

Inspection

Product or documentation
examined

Analysis

Reasoned from models and
measurements

The specification is a foundation for design, not a wish list: it carries the data model, the process decomposition, the state models, and a verification method for all 137 numbered requirements.

Thank you — questions welcome

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